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The Ubiquitous Ewens Sampling Formula

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Abstract. Ewens's sampling formula exemplifies the harmony of mathematical theory, statistical application, and scientific discovery. It not only contributes to the foundations of evolutionary models, but also the neutral theory of biodiversity, Bayesian nonparametric stochastic processes, and inductive inference but also emphasizes its far-reaching influence throughout statistics. We highlight these and many other consequences of Ewens's formula.

Key words and phrases: Ewens's sampling formula, Poisson process, random partition, coalescent process, inductive changeability, logarithmic combinatorial structures, Chinese restaurant process, Ewens–Pitman distribution, Dirichlet process, stick breaking, permanent partition model, cyclic product distribution, clustering, Bayesian nonparametrics, α -permanent.

“It is fair to say that Fisher almost discovered the Ewens sampling formula. He had it in his grasp. After all, he had only to compute the conditional distribution given the sample size, a task that was both statistically natural and, for him, mathematically trivial.”



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Two Early Contributions to the Ewens Saga

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Abstract. The mixture model devised by Fisher, Corbet and Williams [*Journal of Animal Ecology* 12 (1943) 42–58] for species sampling and the sequential prediction approach pioneered by Good [*Biometrika* 40 (1953) 237–264] and Good and Toulmin [*Biometrika* 43 (1956) 45–63] are both closely related to the Ewens sampling formula. Fisher's two-parameter joint distribution for the species counts includes the Ewens distribution as the conditional distribution given the sample size. The log-series model, as it is known in the ecological literature, is closely related to a Poisson process model devised by Arratia, Barbour and Tavaré [*Ann. Appl. Probab.* 2 (1992) 519–535]. Oddly, despite its advantages for statistical inference, Fisher does not mention the conditional distribution. Likewise, although Good (1953) pioneered the sequential prediction approach, neither he nor Toulmin discovered the Ewens process in a form equivalent to the modern-day Chinese restaurant process.

Key words and phrases: Chinese restaurant process, Poisson process, species richness, species sampling.

